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THE COLLECTED
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OF

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THE present volume contains 64 papers numbered 159 to 222 originally published in the years 1857 to 1862: the chronological order is slightly departed from for the sake of including a series of papers in the Memoirs and the Monthly Notices of the Royal Astronomical Society; there are thus some earlier papers which have been left over for the next volume.

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159.

ON SOME INTEGRAL TRANSFORMATIONS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 4–6.]

SUPPOSE that x, a, b, c and x', a', b', c' have the same anharmonic ratios, or what is the same thing, let these quantities satisfy the equation

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ x, & a, & b, & c \\ x', & a', & b', & c' \\ xx', & aa', & bb', & cc' \end{vmatrix} = 0;$$

this equation may be represented under a variety of different forms, which are obtained without difficulty; thus, if for shortness

$$K = a(b' - c')(x' - a') + b(c' - a')(x' - b') + c(a' - b')(x' - c'),$$

then

$$\begin{aligned} Kx &= -\{bc(b - c)(x' - a') + ca(c - a)(x' - b') + ab(a - b)(x' - c')\}, \\ K(x - a) &= (c - a)(a - b)(b' - c')(x' - a'), \\ K(x - b) &= (a - b)(b - c)(c' - a')(x' - b'), \\ K(x - c) &= (b - c)(c - a)(a' - b')(x' - c'). \end{aligned}$$

Consider x, x' as variables; then

$$K^2 dx = (b - c)(c - a)(a - b)(b' - c')(c' - a')(a' - b') dx';$$

let d, d' be any corresponding values of x, x' ; then

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ a, & b, & c, & d \\ a', & b', & c', & d' \\ aa', & bb', & cc', & dd' \end{vmatrix} = 0$$

and we have

$$K(x-d) = D(x'-d');$$

where

$$D = (c' - a')(a' - b')(b' - c') \Lambda,$$

and

$$\Lambda = \frac{(a-d)(b-c)}{(a'-d')(b'-c')} = \frac{(b-d)(c-a)}{(b'-d')(c'-a')} = \frac{(c-d)(a-b)}{(c'-d')(a'-b')}.$$

Suppose $\alpha + \beta + \gamma + \delta = -2$; then

$$(x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta dx = J (x'-a')^\alpha (x'-b')^\beta (x'-c')^\gamma (x'-d')^\delta dx',$$

where

$$J = (b-c)^{\beta+\gamma+1} (c-a)^{\gamma+\alpha+1} (a-b)^{\alpha+\beta+1} (b'-c')^{\beta+\gamma+1} (c'-a')^{\gamma+\alpha+1} (a'-b')^{\alpha+\beta+1} D^\delta.$$

We may in particular take for a', b', c', d' the systems b, a, d, c ; c, d, a, b and d, c, b, a respectively; this gives, writing successively y, z, w instead of x ,

$$\begin{aligned} (x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta dx &= M (y-a)^\beta (y-b)^\alpha (y-c)^\delta (y-d)^\gamma dy \\ &= N (z-a)^\gamma (z-b)^\delta (z-c)^\alpha (z-d)^\beta dz \\ &= P (w-a)^\delta (w-b)^\gamma (w-c)^\beta (w-d)^\alpha dw, \end{aligned}$$

where

$$\begin{aligned} M &= (a-b)^{\alpha+\beta+1} (c-d)^{\gamma+\delta+1}, \\ N &= (-)^{\alpha+1} (a-c)^{\alpha+\gamma+1} (b-d)^{\beta+\delta+1}, \\ P &= (a-b)^{\alpha+\beta+1} (a-c)^{\alpha+\gamma+1} (b-d)^{\beta+\delta+1} (c-d)^{\gamma+\delta+1}; \end{aligned}$$

the relations between the variables x, y, z, w being

$$\begin{aligned} x &= \frac{(c+d)ab - (a+b)cd - (ab-cd)y}{ab-cd - (a+b-c-d)y} \\ &= \frac{(b+d)ac - (a+c)bd - (ac-bd)z}{ac-bd - (a+c-b-d)z} \\ &= \frac{(a+d)bc - (b+c)ad - (ad-bc)w}{ad-bc - (a+d-b-c)w}. \end{aligned}$$

These are, in fact, the formulæ in my note, “On an Integral Transformation,” *Camb. and Dubl. Math. Jour.* t. III. (1848), p. 286 [62], which was suggested to me by Gudermann’s transformation for elliptic functions, (*Crelle*, t. XXIII. (1846), p. 330).

Suppose now that the values of a' , b' , c' , d' are 0, 1, ∞ , ζ , we have in this case

$$x = \frac{a(b-c) + c(a-b)y}{(b-c) + (a-b)y},$$

and representing the denominator by K , then

$$\begin{aligned} K(x-a) &= -(a-b)(a-c)y, \\ K(x-b) &= (a-b)(b-c)(1-y), \\ K(x-c) &= (a-c)(b-c), \\ K(x-d) &= (a-b)(c-d)(y-\zeta), \end{aligned}$$

where

$$\zeta = \frac{(a-d)(b-c)}{(a-b)(c-d)},$$

and we have

$$K^2 dx = (a-b)(a-c)(b-c) dy,$$

whence

$$\begin{aligned} & (x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta dx \\ &= -(-)^\alpha (a-b)^{\alpha+\beta+\delta+1} (a-c)^{\alpha+\gamma+1} (b-c)^{\beta+\gamma+1} (c-d)^\delta y^\alpha (1-y)^\beta (y-\zeta)^\delta dy. \end{aligned}$$

It is easy, by means of this equation, to generalise a remarkable formula given by M. Serret in his memoir, “Sur la Représentation géométrique des Fonctions elliptiques et ultra-elliptiques,” *Liouville*, t. XI. and XII. [1846 and 1847], and *Recueil des Savans étrangers*, t. XI. [1851]¹. In fact, suppose that the indices $\alpha, \beta, \gamma, \delta$ are integers, and that two of these indices, e.g. γ, δ , are negative, the remaining two indices being positive, then writing $-\gamma, -\delta$ instead of γ, δ the integral

$$\int \frac{(x-a)^\alpha (x-b)^\beta dx}{(x-c)^\gamma (x-d)^\delta},$$

where $\gamma + \delta = \alpha + \beta + 2$, depends on the integral

$$\int \frac{y^\alpha (1-y)^\beta dy}{(y-\zeta)^\delta}.$$

Suppose that the fraction under the integral sign is resolved into simple fractions, each of these fractions will be integrable algebraically, except the

¹M. Serret has reproduced the theorem in his very interesting and instructive treatise, “Cours d’Algèbre supérieure,” deuxième édition, Paris, 1854, [quatrième édition, Paris, 1877].

fraction having for its denominator the simple power $y - \zeta$, the integral of which is a logarithm. The coefficient of this fraction is at once found by writing in the numerator $\zeta + (y - \zeta)$ for y ; and expanding in ascending powers of $y - \zeta$ and equating this coefficient to zero, we have

$$\left(\frac{d}{d\zeta}\right)^{\delta-1} \zeta^\alpha (1 - \zeta)^\beta = 0,$$

which [observing that $(\gamma - 1) + (\delta - 1) = \alpha + \beta$] is easily seen to be equivalent to

$$\left(\frac{d}{d\zeta}\right)^{\gamma-1} \zeta^\beta (1 - \zeta)^\alpha = 0.$$

1–2